Program : **Diploma in Polymer Technology**	
Course Code : **5098**	Course Title: **CAD Lab II**
Semester : **5**	Credits: **1.5**
Course Category: **Program Elective**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To provide hands on training on 3D drawing using AutoCAD software.
- To equip the student to be competent in industries for creation and modification of 3D drawings.
- To create an easy platform for students in mastering similar graphics softwares

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Mathematics		Mathematics I	1
Orthographic projections – Pictorial & Multi view		Engineering Graphics	1
AutoCAD 2D drawing skill		CAD Lab - I	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Comprehend opening a various commands and operations required in AutoCAD for drawing 3D objects	12	Applying
CO2	Design and draft various polymer products using AutoCAD 3D	16	Applying
CO3	Develop and draft various moulds for polymer products using AutoCAD 3D	15	Applying
	Lab Exam	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3		3				
CO3	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Comprehend opening a various commands and operations required in AutoCAD for drawing 3D objects**		
M1.01	Comprehend 3D Drawing in AutoCAD – Opening a new file, Modeling in Draw menu for creating particualr objects, 3D draw tool pallet, 3D operation, Solid editing & Surface editing in Modify menu, 3D Views, Visual styles & Orbit in View menu, Multi-view setup etc.	8	Understanding
M1.02	Drawing simple figures like stepped blocks to practice the 3D commands	4	Applying
CO2	**Design and draft various polymer products using AutoCAD**		
M2.01	Design and drafting of simple polymer products like Tensile slab, Paper weight, Eraser, Dumb bell sample, Angular tear samples etc.	6	Applying
M2.02	Design and drafting of complex polymer products like Door mat (Stud, Pyramid & Honey comb designs)	6	Applying
M2.03	Design and drafting of Blow moulded bottle	4	Applying
	Lab exam I	1	
CO3	**Develop and draft various moulds for polymer products using AutoCAD 3D**		
M 3.01	Design and drafting of moulds for polymer products like Tensile slab, Paper weight, Eraser.	5	Applying
M 3.02	Design and drafting of cutting dies for Dumb bell & Angular test samples	4	Applying
M 3.03	Design and drafting of moulds for Door mat (Stud, Pyramid & Honey comb designs) and	6	Applying

	blow moulded bottle		
	Lab exam II	1`	

Text / Reference:

T/R	Book Title/Author
R1	AutoCAD and AutoCAD LT Bible – Ellen Finkelstein (Publisher – John Wiley& Sons Inc.)
R2	Tutorial Guide to AutoCAD – Shawna Lockhart (Publisher – SDC Publications)
R3	Engineering Graphics Essential with AutoCAD instruction – Kirstie Plantenberg (Publisher – SDC Publications)

Online Resources:

Sl.No	Website Link
1	https://leammech.com
2	www.allitebooks.ing
3	www.allaboutfreebooks.com